

Ayush Jain

Research Scientist
Applied Reinforcement Learning, Meta

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EDUCATION

University of Southern California PhD in Computer Science (Advisors: Joseph J. Lim & Erdem Bıyık) GPA: 4.0/4.0	Dec 2024
University of Southern California MS in Computer Science GPA: 4.0/4.0	May 2024
Indian Institute of Technology Delhi B.Tech in Electrical Engineering GPA: 8.99/10	June 2016

RESEARCH INTEREST & EXPERIENCE: REINFORCEMENT LEARNING

PhD Thesis: Decision Making in Complex Action Spaces

I build reinforcement learning algorithms for agents that operate in large, unseen, and complex action spaces, with applications in robotics, recommender systems, and language-model reasoning.

PUBLICATIONS

LLM

- [**ICML 2026**] A. Samanta, A. Magesh, **A. Jain**, K. Asadi, Y. Yu, D. Jiang, B. Vidolov, K. Hassani, P. Sajda, J. Bhandari, Y. Efroni. “Structure Enables Effective Self-Localization of Errors in LLMs”. *International Conference on Machine Learning*.
- [**EACL 2026**] R. Wu, A. Samanta, **A. Jain**, S. Fujimoto, J. Kwon, B. Kretzu, Y. Yu, K. Hassani, B. Vidolov, Y. Efroni. “Imbalanced Gradients in RL Post-Training of Multi-Task LLMs”. *19th Conference of the European Chapter of the Association for Computational Linguistics*.
- [ArXiv] A. Samanta, A. Magesh, **A. Jain**, Y. Yu, D. Jiang, K. Asadi, K. Hassani, P. Sajda, J. Bhandari, Y. Efroni. “Credit Assignment with Resets in Language Model Reasoning”.
- [ArXiv] A. Samanta, A. Magesh, R. Wu, **A. Jain**, Y. Yu, D. Jiang, B. Vidolov, P. Sajda, Y. Efroni, K. Hassani. “Self-Improvement of Language Models by Post-Training on Multi-Agent Debate”.

Robotics

- [**ICLR 2026**] X. Li, **A. Jain**, Z. Yang, Y. Korkmaz, E. Bıyık. “When a Robot is More Capable than a Human: Learning from Constrained Demonstrators”. *International Conference on Learning Representations*.
- [**ICLR 2025**] G. Zhang*, **A. Jain***, I. Hwang, S. Sun, J. Lim. “QMP: Q-switch Mixture of Policies for Multi-Task Behavior Sharing”. *International Conference on Learning Representations*.

RL Algorithms

- [**NeurIPS 2025**] Y. Korkmaz*, U. Bhuwania*, **A. Jain**[†], E. Bıyık[†]. “Actor-Free Continuous Control via Structurally Maximizable Q-Functions”. *Neural Information Processing Systems*.

[RLC 2025] **A. Jain**, N. Kosaka, X. Li, K. Kim, E. Bıyık, J. Lim. “Mitigating Suboptimality of Deterministic Policy Gradients in Complex Q-functions”. *Reinforcement Learning Conference*.

Outstanding Paper Award on Empirical Reinforcement Learning Research

[ICLR 2022] **A. Jain***, N. Kosaka*, K. Kim, J. Lim. “Know Your Action Set: Learning Action Relations for Reinforcement Learning”. *International Conference on Learning Representations*.

[ICML 2020] **A. Jain***, A. Szot*, and J. Lim. “Generalization to New Actions in Reinforcement Learning”. *International Conference on Machine Learning*.

Earlier Work

[COLING 2018 Workshop] **A. Jain**, V. Singh, S. Ranjan, R. Rajkumar, S. Agarwal. “Uniform Information Density Effects on Syntactic Choice in Hindi”. *Workshop on Linguistic Complexity and NLP*.

[DGfS 2017] **A. Jain**, V. Singh, S. Agarwal, and R. Rajkumar. “Uniform Information Density models for language production”. *39th Annual Conference of the German Linguistic Society, DGfS 2017*.

INDUSTRY EXPERIENCE

Meta, Sunnyvale, USA Feb 2025 – Present

Research Scientist in Applied Reinforcement Learning Team, Monetization Org

→ Reinforcement learning for optimizing ad bidding and auction as a contextual bandit.

→ Reinforcement learning to optimize the recommendation of ads to users on Meta Platforms.

→ RL post-training and evaluation of multi-turn LLM conversation assistants.

Meta Reality Labs Research, Redmond, USA May 2024 – Dec 2024

Research Intern with Nitin Kamra

→ Reinforcement learning based virtual agents for instruction-following in **Android** devices.

Microsoft Research, Montreal, Canada May 2023 – Jan 2024

Research Intern with Eric Yuan and Marc-Alexandre Côté

→ Combine discrete prompt optimization in large language models (LLM) with gradient-optimization in neural networks for natural-language reasoning tasks and instruction-following **Minecraft** agents.

Naver AI Research, Seoul, South Korea (remote) May 2022 - Aug 2022

Research Intern with Kyung-Min Kim (Naver CLOVA), Joseph J Lim June 2021 – Dec 2021

→ Reinforcement Learning in **recommender systems** for large and varying action spaces, like streaming news recommendations, using graph attention networks to address varying listwise slate-actions.

Samsung Research, Seoul, South Korea Sep 2016 – June 2018

Engineer at Data Analytics Lab with James Geraci, Yunsu Lee

→ Market share prediction and data unification for various Samsung appliances using time series models.

→ Failure prediction with time-series anomaly detection and information extraction from text modeling.

Samsung Electronics, Suwon, South Korea May 2015 – July 2015

Software Engineer Intern with Sungmok Seo

→ Design of smart thermostat algorithm that learns and adapts to user schedules.

PATENT

J. Lee, M. Kim, **A. Jain**, T. Hwang, J. Kim, H. Cho. “Method and Apparatus for Managing Operation Data of Appliance for Failure Prediction”. U.S. Patent No. 11,182,235. 11/23/2021.

TEACHING & REVIEWER SERVICES

Teaching Assistant, University of Southern California (USC)

Course: Deep Learning and its Applications, CSCI-566

Joseph J. Lim: Fall 2019, Sprint 2019, Fall 2020

Jesse Thomason: Spring 2023

Yue Zhao: Spring 2024

Yan Liu: Fall 2024

Reviewer Services:

International Conference on Learning Representations (ICLR): 2023, 2024, 2025, 2026

Conference on Neural Information Processing Systems (NeurIPS): 2023, 2024, 2025, 2026

International Conference on Machine Learning (ICML): 2025, 2026

Conference on Robot Learning (CoRL): 2021, 2022, 2023, 2024

Reinforcement Learning Conference (RLC): 2025, 2026

Association for the Advancement of Artificial Intelligence (AAAI): 2026

ACHIEVEMENTS

Outstanding Paper Award on Empirical Reinforcement Learning Research at Reinforcement Learning Conference (2025)

Director’s Award for ranking in the top 7% at IIT Delhi (2013-14).

All India Rank of 198 in IITJEE (2012).

All India Rank of 91 in AIEEE (2012).

Central Board of Secondary Education Merit Certificate for securing top 0.1% (2012).

KVPY Fellowship Award and National Talent Search Examination (NTSE) Scholarship Award.

All India Rank of 3 in the National Science Talent Search Examination (NSTSE) (2011).

Top 1% nationally in Physics, Chemistry, and Astronomy (2011).

Top 30 in the Indian National Astronomy Olympiad (2012).